

Ismet Bilgic

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EDUCATION

Bachelor of Applied Science, Software Engineering (CO-OP)

Expected April 2027

University of Ottawa, Ottawa, ON

- Awarded Chancellor's Scholarship | GPA: 3.94/4

WORK EXPERIENCE

Data Engineer Intern

September 2024-Present

Bombardier Aerospace, Dorval, Quebec

- Built a production real-time monitoring service (Python, SQLAlchemy, APScheduler) tracking a 480-aircraft fleet via a per-tail state machine with adaptive polling, circuit breakers, and automatic crash recovery, now being deployed on Snowflake to continuously record JetNet API flight data; backed by 118 automated tests and Alembic-versioned migrations
- Built a three-source validation engine (JetNet vs internal Snowflake/AHMS vs FlightAware) to assess the feed before adoption, surfacing flights for 17 aircraft missing from internal systems (+3.7% coverage) and diagnosing a GPS vs barometric altitude mismatch that lifted cruise agreement from under 20% to 95%
- Developed a Python automation tool using FFT cross-correlation and multithreaded processing to validate signal lag between two flight recording systems across 1,400+ parameters for new software version certification, reducing a weeks-long manual process requiring 3-4 engineers to under an hour
- Engineered two Snowflake views (300+ lines each) for Global 7500 and retrofit fleet watchlists, automating daily Power BI reporting via key-pair authentication, eliminating 1-1.5 days of weekly manual work for 50-60+ stakeholders across maintenance, AHMS, and engineering
- Built an end-to-end Python data pipeline (validation, cleaning, processing, enrichment) to calculate monthly aircraft data transfer KPIs across Global 7500 and retrofit fleets, integrating Snowflake, AWS S3, and DSP reporting sources, reducing a 2-hour manual process to 30 minutes

Deputy Coding Team Lead

October 2024-June 2025

Scale AI, Remote

- Served as a deputy lead within a ~1,000-contributor Turkish LLM-training pod (one of 15 in the program), coordinating reviewer feedback, quality standards, and escalation workflows while maintaining a 4.4/5 average client satisfaction score
- Earned top-tier 'Oracle' contributor status on the platform, reviewing and quality-controlling 250+ coding tasks across Python, Java, SQL, and JavaScript to improve model accuracy and localization across English, French, and Turkish

IT Operations Specialist Intern

January 2024-April 2024

Deloitte Touche Tohmatsu Limited, Ottawa, Ontario

- Engineered Python solutions (pandas, NumPy) to categorize 47 types of stuck manifests and automate troubleshooting, reducing issue resolution from 45 minutes to 3-5 minutes
- Developed PowerShell and Python automation, including Selenium-based tools for pre-patch web verification and Excel data transfers, saving the team 20% in man-hours over four months

Program Analyst Intern

May 2023-September 2023

Bombardier Aerospace, Dorval, Quebec

- Built a Python (openpyxl) tool that auto-populated a 20-milestone-per-PCR tracking matrix from the source sheet, syncing completion, pushed, and cancelled statuses with dates, replacing a manual update run before and after each weekly program review meeting
- Developed a Python script to scan Lotus Notes across ~150 PCRs, automatically detecting and flagging obsolete records left open past closure to improve project-tracking accuracy and data integrity

PROJECTS

Echo Explorer | Java, SQL, Google Maps API

- Developed an Android navigation application (Java, SQL) tailored for accessibility, designed for visually impaired users
- Integrated the Google Maps API to provide real-time navigation assistance

SKILLS

Languages: Java, Python, SQL, HTML/CSS, JavaScript, Bash, PowerShell

Frameworks/Libraries: pandas, NumPy, openpyxl, Selenium, SQLAlchemy, Alembic, APScheduler, Playwright, pytest

Databases/Cloud: PostgreSQL, MySQL, Snowflake, AWS S3, AWS Lambda

Tools: Power BI, Git, SSMS